

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ

Chapter 14: Equation of a Line

Previous year board questions, topic-wise: 3 questions

The tag after each question gives the board examination year and the marks. Solutions are in a separate PDF with the same question numbers.

1 Forms of the equation

1. $ABCD$ is a parallelogram with vertices $A(3, 3)$, $B(0, -2)$, $C(-4, -2)$ and $D(-1, 3)$, read from a graph. (a) Find the point P where the diagonals meet. (b) Find the slopes of CB and DA and check that these sides are parallel. (c) Find the equation of the diagonal AC .

ICSE 2025, 5 marks

2 Lines parallel or perpendicular to a given line

2. Which line passes through the origin and is parallel to $3x + 4y + 7 = 0$? (a) $3x + 4y + 5 = 0$ (b) $4x - 3y - 5 = 0$ (c) $4x - 3y = 0$ (d) $3x + 4y = 0$ ICSE 2025, 1 mark
3. The line PQ meets the positive x -axis at P , 5 units from the origin, and the negative y -axis at Q , 3 units from the origin. The line RS passes through the origin and is perpendicular to PQ . Find (a) the coordinates of P and Q (b) the equation of RS . ICSE 2025, 3 marks

These questions are based on questions asked in past board examinations. They have been reworded by MathSetu, the figures are redrawn, and the solutions are MathSetu's own. MathSetu is an independent study resource and is not affiliated with or endorsed by CBSE, CISCE, NCERT or any publisher.